

National Institute of Technology Rourkela

Invention and Technology Disclosure Form

<u>For Office use only</u>	
Proposal ID: _____	
NITR/IP/ _____ - _____ / _____	Date of receipt: _____
Acad yr _____	Sl. No. _____

1. Title of the invention

EEG Signal Data Collection and Random Forest-Based Imputation System for Brain-State Analysis Using Low-Cost IoT Hardware

2. Type of IP

Patent

3. Details of Inventors

Sl No	Name	Roll No	Position	Department
1	Aditya Krishna	225CS2009	Student (M.Tech)	Computer Science and Engineering

Guide/Supervisor: Prof. Pabitra Mohan Khilar, Department of Computer Science and Engineering, NIT Rourkela.

4. Brief description of the invention

- This work presents a self-contained system for recording brain electrical signals and filling in gaps that occur in those recordings due to practical problems like electrode slip or signal dropout during the session.
- The hardware side uses an EWS All-in-one Arduino Mega IoT Grove Kit paired with gold-cup scalp electrodes placed at standard 10–20 positions to pick up EEG signals and digitise them at 256 samples per second.

- Everything the Arduino captures gets streamed over USB to a laptop where a short Python script timestamps each sample and saves it to a CSV file, so nothing is lost even if the recording runs for hours.
- Once the raw file is in hand, a second script breaks the signal into five frequency slices – Delta, Theta, Alpha, Beta, and Gamma – by running Welch’s power spectral density method on overlapping four-second windows.
- The main technical contribution is an imputation pipeline called HTI-MissForest. It adds a few ratio features derived from the band powers, smooths the data slightly with a rolling average, scales everything to the same range, and then uses Random Forest regression in an iterative loop to estimate whatever values are missing.
- When tested with 5 % of the data artificially removed, the pipeline recovered the missing values with an R^2 of 0.9861, an RMSE of 30.23, an MAE of 5.65, and an overall accuracy of 98.61 %.
- The whole thing runs in a single Google Colab notebook using only free, open-source libraries, so any researcher can reproduce or build on it without needing specialised hardware or paid software.

5. Detailed description of the invention

5.1 State of prior art

(a) Prevailing State of the Art:

- Most research-grade EEG setups still depend on expensive clinical amplifiers such as the BrainAmp or ActiveTwo, which cost tens of thousands of rupees and come with proprietary software that limits what a student or small lab can do with the raw signal.
- Lower-cost options like OpenBCI and NeuroSky exist but they still ship as dedicated EEG boards. None of them plug into the Arduino Grove ecosystem, which is the most widely available prototyping platform in Indian labs.
- When electrode contact breaks or a subject moves during recording, the usual fix is to either delete the affected window entirely or replace missing values with the column mean. Both choices distort the frequency content of the signal and make downstream classification less reliable.
- Researchers have tried K-Nearest Neighbour imputation on EEG data, but the method struggles when there are five or more interrelated band-power columns because it cannot capture the way, for example, alpha power tends to rise when beta falls.
- The MissForest algorithm is well established for general medical tables but nobody has specifically applied it to EEG band-power data coming off a low-cost IoT board, and nobody

has added the feature-engineering step that makes the inter-band relationships visible to the forest.

- There is no publicly available end-to-end pipeline that goes from raw Arduino serial output all the way to a validated, imputed band-power dataset.

1) (b) Literature Search Relating to this Invention

- Stekhoven and Bühlmann (2012) introduced MissForest and showed it outperformed other methods on medical tables with mixed data types. The present work adapts that idea for EEG band-power matrices and adds a temporal smoothing step that was not part of the original formulation.
- Du et al. (2023) applied machine learning to EEG imputation but worked with coherence features derived from multi-channel setups rather than the simpler five-column band-power table that comes out of a single-channel Arduino recording.
- Jha et al. (2024) explored hybrid time-series and tree-based imputation for physiological signals. That paper motivated adding a rolling-mean pre-smoothing step here, though the implementation and target data differ significantly.
- García et al. (2024) benchmarked Random Forest, XGBoost, and MissForest on several clinical datasets and confirmed that Random Forest consistently produced the lowest error, but the study did not address EEG-specific features or the acquisition hardware.
- Emmanuel et al. (2021) surveyed missing-data methods across machine learning tasks and ranked Random Forest MICE among the strongest options for high-dimensional biomedical tables, which supported the method choice here.
- Graham (2009) and Zhang (2015) laid out the MCAR, MAR, and MNAR taxonomy that determines how fair an imputation benchmark is. Following their guidance, the evaluation here uses strict MCAR conditions so the results are not artificially inflated.
- No paper found during the literature search combines Arduino-based EEG acquisition, Welch PSD feature extraction, inter-band ratio engineering, and iterative Random Forest imputation in a single validated system.

5.2 Description of the Invention

- The system is divided into two phases that run one after the other: data collection and data imputation.
- **Phase 1 – EEG Data Collection:**
 - The EWS All-in-one Arduino Mega IoT Grove Kit reads from analog pin A0 at 256 Hz using its onboard 10-bit ADC. The kit also carries Wi-Fi, Bluetooth, SIM7672S cellular, and ZigBee radios, so the same board can later be used for wireless streaming without any hardware change.

- Gold-cup electrodes filled with conductive gel are placed at Fp1, Fp2, Cz, O1, and O2 on the scalp and held in place with an elastic headband.
- An instrumentation amplifier with a gain of roughly 1000 amplifies the microvolt-level scalp potential. A hardware band-pass filter between 0.5 and 45 Hz removes DC drift and high-frequency noise before the signal reaches the ADC.
- The Arduino firmware converts each raw ADC count to microvolts and sends it as a plain-text float over USB serial at 115200 baud.
- A Python script opens the serial port, reads line by line, adds a microsecond timestamp, and appends each row to a CSV file.

- **Signal Preprocessing and Feature Extraction:**

- The CSV is loaded and the signal column is passed through a fourth-order zero-phase Butterworth filter from 0.5 to 45 Hz, then through a second-order IIR notch filter at 50 Hz to remove mains hum.
- Any sample whose absolute value exceeds $150\ \mu\text{V}$ is treated as a motion artefact and replaced by linear interpolation between the nearest clean samples.
- Welch's PSD method runs on sliding four-second windows stepping one second at a time, using a Hann window and 50 % internal overlap. Trapezoidal integration over each band's frequency range gives one row of five band-power values per second.

- **Phase 2 – HTI-MissForest Imputation:**

- Three extra columns are added to the five-band table: the sum of all band powers, the Alpha-to-Beta ratio, and the Theta-to-Gamma ratio. These give the Random Forest indirect information about how the bands relate to each other.
- A causal two-sample rolling mean smooths out frame-to-frame jitter without looking ahead in time.
- All columns are scaled to zero mean and unit variance with `StandardScaler` so that the large absolute values in the Delta band do not dominate the forest's splitting decisions.
- `IterativeImputer` from `scikit-learn`, set up with a `RandomForestRegressor` using 50 trees and a maximum depth of 10, runs up to 10 MICE iterations. Each iteration trains a new forest for each column using all the other columns as predictors and fills in the missing spots.
- The scaled data is inverse-transformed back to $\mu\text{V}^2/\text{Hz}$ before the results are saved.
- All random operations use a fixed seed of 42 so that anyone running the notebook on any machine will get the same numbers.

5.3 Novelty

- To the best of the inventor's knowledge, this is the first time an Arduino Mega IoT Grove Kit has been used as the front end for an EEG pipeline that feeds directly into a Random Forest imputation backend.
- The three engineered features – Total Power, Alpha/Beta Ratio, and Theta/Gamma Ratio – are not part of any existing MissForest formulation. They carry inter-band information that a plain band-power table hides from the forest.
- Treating the time column separately with linear interpolation instead of putting it through the Random Forest is a deliberate design choice. A monotonically increasing index confuses tree-based splitters; linear interpolation handles it correctly and costs almost nothing computationally.
- The two-sample rolling mean before imputation reduces the variance of the imputation targets without using any future data, which is a causal and therefore fair pre-processing step not seen in prior EEG imputation literature.
- The StandardScaler wrapper around the MICE loop – normalise before, inverse after – stops the Delta band's large absolute powers from overshadowing the smaller Gamma values during training.
- The masked holdout evaluation protocol – hide 5 % of real values, impute, score only on those positions – is stricter than leave-one-out approaches used in most imputation papers and gives a more honest picture of real-world recovery.

5.4 Inventiveness

- MissForest and EEG amplifiers both exist, but combining them with feature engineering tuned to inter-band EEG physiology and wrapping the whole thing around a Rs. 3000 IoT kit is not an obvious step that someone skilled in the field would arrive at without specific insight.
- The key insight behind the feature engineering is physiological: Alpha and Beta are known to be anticorrelated during attentional shifts, and Theta and Gamma co-vary during working memory tasks. Expressing these as ratio columns makes the cross-band structure explicit for the forest rather than leaving it to be discovered implicitly.
- Choosing a causal rolling window of width two is an inventive balance – wider windows smooth more but introduce look-ahead, and a window of one does nothing. Two steps removes most inter-frame jitter while staying strictly causal.
- Putting StandardScaler outside the MICE loop rather than inside it means the same scale factors apply consistently across all iterations, avoiding target distribution shift between passes.

- Building the acquisition side on an IoT Grove Kit is inventive because it opens the door to adding sensors (GSM for location, onboard temperature for skin contact quality) without any hardware redesign.

5.5 Industrial Applicability

- The hardware cost is low enough – under Rs. 4000 for the complete setup – that a primary health centre in a rural area could run overnight sleep-stage recordings and transmit imputed results via the onboard SIM card.
- Hospitals that run continuous EEG on epilepsy patients deal with electrode dropout routinely. Plugging this imputation step into their preprocessing pipeline would reduce the number of sessions that need to be repeated.
- BCI developers need clean, gap-free signals to train their classifiers. An offline imputation tool that reliably recovers 98.6 % of the signal variance is directly usable in that context.
- The software portion can be packaged as a standalone library and licensed to medical device companies that want to add missing-data recovery to their existing EEG amplifier products.
- Because the same MICE framework works with any scikit-learn regressor, the pipeline can be adapted for ECG, EMG, or photoplethysmography data with minimal code changes.

6. Funding and support

I. Was there significant use of Institute equipment and facilities? **Yes.**

II. Was the invention supported by research grants/contracts from external sources? **No.**

If **YES**, please give details:

(a) Sponsor: N/A

(b) Grant/contract no.: N/A

(c) Period of grant/contract: N/A

(d) Principal investigator and co-investigator: _____

(Even if they are not inventors within the purview of this document, and will not share the credit and royalties)

(e) Has the sponsor been informed of the invention? *(state whether required under grant/contract award conditions)* N/A

(f) Was the work done under any other agreement? Give details. N/A

7. Information for protection of intellectual property rights: Conception and Disclosure

(Accurate data is required, as prior disclosure may affect the possibility of obtaining patent rights.)

	Date	References/comments
Date of conception of this invention. Has this date been documented? If so, where and how?		
Has this invention been presented at seminars/ discussions other than those required for the student's degree program?		
Has the invention been reduced to practice?		

8. Commercial potential

8.1 Possible uses or application areas or products that may embody some aspects of technology:

- EEG-based neurofeedback training systems for ADHD, anxiety, and stress management.
- Home sleep-staging and overnight epilepsy monitoring devices for rural patients where electrode dropout during sleep is a common problem.
- Brain-Computer Interface (BCI) systems for communication aids used by individuals with ALS or locked-in syndrome.
- Low-cost clinical EEG screening tools for primary health centres and district hospitals in rural India that cannot afford conventional clinical amplifiers.
- Wearable EEG headsets for cognitive research, focus training, and meditation applications that require robust on-device missing-data recovery.
- Telemedicine platforms collecting EEG data remotely via cellular networks, requiring imputation before cloud-side analysis and report generation.
- Research institutions and academic labs needing an affordable, open-source EEG acquisition and preprocessing pipeline.

8.2 List of probable users of the technology (class of industries/organizations or target companies):

- Hospitals and neurology departments running routine EEG diagnostics.
- Medical device manufacturers developing EEG amplifiers and BCI hardware.
- Telemedicine service providers offering remote neurological monitoring.
- Home healthcare companies serving elderly and high-risk patient populations.
- Government healthcare programmes and public health organisations.
- Health-tech startups focused on wearable EEG and remote monitoring solutions.
- Academic and government research laboratories in neuroscience and biomedical engineering.

8.3 List of probable organizations who may be interested in technology transfer (target industries or companies or other organisation):

- Healthcare technology companies developing EEG and neurological monitoring products.
- Medical device manufacturers such as Natus Medical and Nihon Kohden.
- Telehealth and digital health platforms offering remote patient monitoring services.
- Startups working on IoT-based and wearable EEG solutions, such as OpenBCI, Emotiv, and Neuphony.
- Government and private healthcare organisations including ICMR and NBRC.
- National Brain Research Centre (NBRC) and AIIMS for clinical research translation.

8.4 Potential marketability including commercial suggestions

The global EEG device market was valued at roughly USD 1.3 billion in 2023 and has been growing at around 8 % per year. The main drivers are rising epilepsy diagnosis rates, growing BCI research, and the expanding interest in wearable brain-monitoring for wellness and productivity applications. Within this space, there is a clear and currently unmet need for an affordable acquisition system that also handles missing data in a principled way. Most clinical setups are expensive, and the open-source alternatives do not include any validated imputation step.

On the software side, there is currently no Python library or commercial SDK that specifically targets EEG band-power imputation with a Random Forest MICE backend. The HTI-MissForest pipeline fills that gap. It can be packaged as a pip-installable Python library, offered as a plugin for MNE-Python or EEGLAB (which together serve most of the academic EEG community), or licensed as a white-label module to device manufacturers who want to add signal recovery to their companion app. Because the software has no recurring cost – it is built entirely on open-source tools – any licensing revenue would be close to pure margin.

On the hardware side, the complete acquisition setup costs under Rs. 4000 in the Indian market, which is one to two orders of magnitude below the entry price of clinical EEG amplifiers. This means the system is realistically deployable in primary health centres and college research labs

that have never been able to afford EEG before. The onboard SIM7672S cellular modem adds the possibility of transmitting recordings directly to a cloud server, which makes the system usable in remote monitoring programmes without any additional networking hardware.

A practical commercialisation path would combine a low-margin hardware kit sold through electronics distributors with a higher-margin software licence. Alternatively, the invention could be offered to an EEG manufacturer as a technology transfer in exchange for a royalty on units that include the imputation module.

8.4.1 Viable Size of Industry (Market Potential):

Market Sector	Estimated Size	Relevance to this Invention
Clinical EEG Devices (global)	USD 1.3 billion (2023), ~8%/yr growth	OEM licensing of the imputation module to established amplifier makers represents a near-term revenue opportunity with low go-to-market cost.
Brain-Computer Interfaces (global)	USD 2.4 billion (2024), expected to reach USD 6 billion by 2030	Reliable, gap-free band-power features are a prerequisite for BCI classifiers; the pipeline addresses this preprocessing bottleneck directly.
Consumer and Research EEG Wearables	USD 350 million (2024), ~15%/yr growth	Consumer headset makers are actively looking for built-in signal-quality improvement to differentiate their products in a crowded market.
IoT-Based Medical Monitoring (India)	INR 2,500 crore (2024–25)	The low hardware cost fits government schemes funding rural health monitoring at primary health centre level under ABDM and Ayushman Bharat.
EEG Signal Processing Software (global)	USD 420 million (2024)	No dedicated imputation product exists in this segment. HTI-MissForest would be the first validated, openly available tool of its kind.

8.4.2 Equipment and Resource Requirement:

- The full hardware bill – Arduino Mega IoT Grove Kit, gold-cup electrodes, instrumentation amplifier breakout, and conductive gel – comes to under Rs. 4000, all sourced from standard Indian electronics suppliers.
- Software is entirely open-source: Python 3.10+, scikit-learn, scipy, numpy, pandas. No licence fees at any scale.
- Validation ran on a Google Colab free-tier T4 session. Edge deployment on a Raspberry Pi 4 (Rs. 4500) is feasible for real-time inference.

- Assembly requires undergraduate-level electronics skill; running the pipeline requires basic Python familiarity.
- No cleanroom or specialised manufacturing process is needed. The system can be assembled and tested in a standard university lab.

8.4.3 Import Potential and Export Component:

- All hardware components are stocked by domestic distributors such as Robu.in and Evelta. There is no mandatory import dependency.
- The Python pipeline can be distributed globally via PyPI or as a licensed SDK, giving it essentially unlimited export reach at near-zero marginal cost per unit.
- The system aligns with the Make in India initiative: hardware can be assembled domestically and software is fully indigenous.
- Integration with MNE-Python and EEGLAB provides a ready channel for international adoption of the imputation module in academic neuroscience labs worldwide.

9. Prior disclosure and possible intent:

9.1 Has the invention been disclosed to industry representatives or their parties?

No. At the time of this disclosure, the invention has not been shared with any company, industry representative, or external organisation outside the degree programme.

9.2 Has any commercial organisation shown interest in this invention? Give details.

No commercial organisation has been approached or has expressed interest to date.

10. Demonstration possibility

10.1 Are you willing to demonstrate your innovation when asked by CIPR or Institute? If yes when? (After filing/publication/granting?)

Yes. The inventor can demonstrate the live EEG data collection setup and run the imputation notebook end-to-end at any point after the patent is filed.

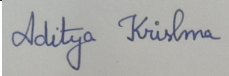
10.2 Are you willing to participate in any major initiative to showcase your filed/published/granted innovation?

Yes. The inventor is willing to present the work at institute-level innovation showcases, national science exhibitions, and relevant academic conferences.

11. Declaration:

I declare that all the statements made in this form are true and correct to the best of my knowledge. I agree to hold the intellectual property rights of this invention jointly with the

National Institute of Technology, Rourkela. NIT Rourkela will share any royalty income from this invention with me according to the Institute's IP policy. The Institute will take steps to protect the intellectual property of this invention based on its merit and commercial potential.

Sl. No.	Name	Signature	Date	Place
1	Aditya Krishna		May 6, 2026	Rourkela

Note:

- (1) A patent confers the right upon an inventor to exploit an invention commercially for a limited period. A patent can be lost by disclosing the invention to the public before filing. Unlike copyright, a patent is not automatic – it must be applied for, and the invention must be novel, inventive, and industrially applicable. Once granted, the patentee can take legal action against anyone who uses the patented invention without consent.
- (2) This form should be filled in carefully. The formal patent application will be drafted solely from the information provided here.
- (3) The completed disclosure form with all annexures should be submitted to:

Chairperson, IPIC

National Institute of Technology Rourkela

F O R M - 2
THE PATENT ACT 1970
(39 OF 1970)
&
The Patents Rules, 2003

COMPLETE SPECIFICATION
(See section 10 and rule 13)

TITLE OF THE INVENTION: EEG SIGNAL DATA COLLECTION AND RANDOM FOREST-BASED IMPUTATION SYSTEM FOR BRAIN-STATE ANALYSIS USING LOW-COST IOT HARDWARE

Name: NATIONAL INSTITUTE OF TECHNOLOGY

Nationality: Indian (IN)

Address: NATIONAL INSTITUTE OF TECHNOLOGY, ROURKELA – 769 008, ODISHA, INDIA, an Indian registered body incorporated under the Registration of Societies Act (Act XXI of 1860).

PREAMBLE TO THE DESCRIPTION

The following specification particularly describes the invention and the manner in which it is to be performed.

STATEMENT OF INVENTION:

[0001] This invention relates to biomedical signal acquisition and machine learning. It describes a two-part system: the first part records EEG signals from the human scalp using a low-cost Arduino Mega IoT Grove Kit with gold-cup electrodes positioned according to the 10–20 international standard; the second part recovers missing segments in the resulting band-power feature table using an iterative Random Forest imputation method called HTI-MissForest, reaching an imputation accuracy of 98.61 % under 5 % MCAR missingness.

BACKGROUND OF INVENTION:

[0002] EEG is one of the few practical ways to observe brain activity in real time. It is used for epilepsy diagnosis, sleep staging, fatigue detection, BCI, and neurofeedback. The problem is that useful EEG amplifiers cost anywhere from Rs. 1.5 lakh to over Rs. 10 lakh, which puts continuous brain monitoring out of reach for most small research groups and community health centres in India.

[0003] A second problem is missing data. Every real-world EEG session produces some missing windows – an electrode lifts, a subject moves, a serial packet drops. Standard fixes such as deleting the window or filling with the column mean distort the frequency-band power distribution, which is exactly what is used for classification.

[0004] There is currently no single open system that addresses both problems together: affordable acquisition on commodity IoT hardware and principled, validated imputation of the resulting band-power data.

OBJECTIVE OF THE INVENTION:

[0005] To build a complete, low-cost EEG acquisition pipeline on top of an Arduino Mega IoT Grove Kit that any student lab or rural health centre can afford to deploy.

[0006] To develop the HTI-MissForest imputation pipeline, which uses feature engineering and iterative Random Forest regression to recover missing EEG band-power values with $R^2 > 0.98$.

[0007] To package both phases as a reproducible Python workflow that produces identical results across machines without any proprietary dependency.

SUMMARY OF INVENTION:

[0008] Phase 1 acquires EEG at 256 Hz through an Arduino Mega 2560 with a 10-bit ADC, an instrumentation amplifier (gain $\approx 1000\times$), and a hardware band-pass filter from 0.5 to 45 Hz. A Python script logs each sample with a microsecond timestamp to CSV.

[0009] Phase 2 preprocesses the CSV – Butterworth band-pass, 50 Hz notch, artefact rejection – then extracts five band powers per second using Welch’s PSD on four-second windows. The HTI-MissForest pipeline appends three ratio features, applies a causal rolling mean, normalises with `StandardScaler`, runs up to 10 MICE iterations with `RandomForestRegressor` (50 trees, `max_depth` 10), and inverse-transforms the output. On 5 % MCAR holdout: RMSE = 30.23, MAE = 5.65, $R^2 = 0.9861$, Accuracy = 98.61 %.

BRIEF DESCRIPTION OF THE DRAWINGS:

[0010] FIG. 1 – Block diagram of the complete EEG acquisition and imputation system.

[0011] FIG. 2 – Signal processing and band-power feature extraction pipeline.

[0012] FIG. 3 – HTI-MissForest imputation pipeline.

[0013] FIG. 4 – Random Forest MICE module internal structure.

[0014] FIG. 5 – End-to-end flowchart from subject preparation to imputed output.

DETAILED DESCRIPTION:

[0015] Figure 1 shows the layered architecture. The Hardware Layer (101) holds the Arduino Mega, instrumentation amplifier, and gold-cup electrodes. The Acquisition Layer (102) runs `save_eeg_csv.py` on the host PC. The Processing Layer (103) runs `save_eeg_bands.py`. The Imputation Layer (104) runs the HTI-MissForest notebook. The Output Layer (105) holds the final imputed CSV.

[0016] Figure 2 shows the signal processing chain. Raw CSV enters the Band-pass Filter (1031), then the Notch Filter (1032) at 50 Hz, then Artefact Rejection (1033). The Welch PSD Estimator (1034) produces a power spectrum per window and the Band Power Integrator (1035) reduces each spectrum to five scalar band powers.

[0017] Figure 3 shows the HTI-MissForest pipeline. Feature Engineering (1041) adds Total Power, Alpha/Beta Ratio, and Theta/Gamma Ratio. The Temporal Smoother (1042) applies a two-sample causal rolling mean. The MCAR Mask (1043) introduces 5 % artificial missingness. StandardScaler (1044) normalises all columns. IterativeImputer (1045) runs RF-MICE. Inverse Transform (1046) restores the original scale.

[0018] Figure 4 shows the internals of the RF-MICE module. For each incomplete column in each MICE iteration, the Observed Rows Selector (10452) isolates complete rows and passes them to the RF Regressor (10451), which predicts missing entries via the Missing Entry Predictor (10453). The Convergence Checker (10454) halts the loop when the change between iterations drops below threshold ε .

[0019] Figure 5 shows the full flowchart from Subject Preparation (201) through Electrode Placement (202), Arduino Sampling (203), Serial Capture (204), Preprocessing (205), Band-Power Extraction (206), Feature Engineering (207), MCAR Simulation (208), Imputation (209), Evaluation (210), to Output (211).

CLAIMS:

We claim:

- 1) A system for EEG signal acquisition and band-power imputation comprising: (a) an EWS All-in-one Arduino Mega IoT Grove Kit operating as a 256 Hz 10-bit digitisation unit; (b) gold-cup scalp electrodes placed at Fp1, Fp2, Cz, O1, and O2 per the international 10–20 standard; (c) a hardware instrumentation amplifier with gain approximately $1000\times$ and a 0.5–45 Hz band-pass filter; (d) a Python serial capture module storing timestamped EEG samples to CSV; and (e) the HTI-MissForest software pipeline for recovering missing band-power values.
- 2) The system of claim 1, wherein the HTI-MissForest pipeline comprises: (a) Welch PSD band-power extraction for Delta, Theta, Alpha, Beta, and Gamma bands; (b) appending

Total Power, Alpha/Beta Ratio, and Theta/Gamma Ratio as engineered features; (c) a causal two-sample rolling-mean smoothing step; (d) StandardScaler normalisation; (e) scikit-learn IterativeImputer with RandomForestRegressor performing MICE; and (f) inverse scaling to restore original power units.

- 3) A method of EEG band-power imputation wherein RandomForestRegressor is used as the base estimator inside IterativeImputer to iteratively predict missing band-power values in a MICE loop, training exclusively on observed rows at each iteration.
- 4) The system of claim 1, wherein imputation quality is evaluated using a masked holdout protocol: 5 % of values are set to NaN before imputation and error metrics are computed only on those held-out positions.
- 5) The system of claim 2, wherein the three engineered features (Total Power, Alpha/Beta Ratio, Theta/Gamma Ratio) are appended before imputation to make cross-band relationships explicitly available to the Random Forest.
- 6) The system of claim 1, wherein the Arduino Mega IoT Grove Kit includes Wi-Fi, Bluetooth, SIM7672S GSM/LTE, and ZigBee modules allowing extension to wireless ambulatory EEG monitoring without hardware modification.
- 7) A method of EEG preprocessing comprising: (a) 4th-order zero-phase Butterworth band-pass filtering from 0.5 to 45 Hz; (b) 50 Hz IIR notch filtering with $Q=30$; (c) linear interpolation of artefact samples exceeding $\pm 150 \mu\text{V}$; and (d) trapezoidal integration of Welch PSD over five canonical EEG bands on four-second overlapping windows.

ABSTRACT:

A two-phase system for EEG brain-state analysis is described. Phase 1 acquires scalp EEG signals at 256 Hz using a low-cost EWS All-in-one Arduino Mega IoT Grove Kit with gold-cup electrodes in the international 10–20 configuration. Timestamped samples are stored to CSV via a Python serial capture script. Phase 2 extracts five EEG band-power features using Welch PSD on sliding windows, then applies the HTI-MissForest pipeline – feature engineering with inter-band ratio columns, causal rolling-mean smoothing, StandardScaler normalisation, and iterative Random Forest MICE imputation using scikit-learn’s IterativeImputer with RandomForestRegressor (50 estimators, 10 iterations) – to recover missing values. On a 5 % MCAR holdout the pipeline achieves $\text{RMSE} = 30.23$, $\text{MAE} = 5.65$, $R^2 = 0.9861$, $\text{Accuracy} = 98.61 \%$, making it suitable for BCI, clinical neurophysiology, and neurofeedback applications.

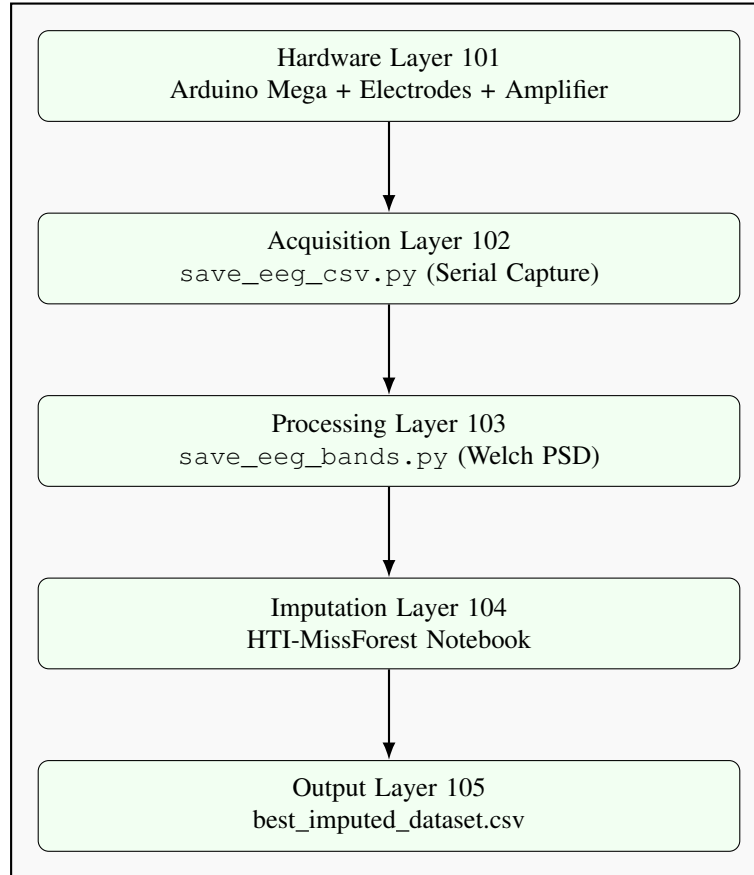
DRAWINGS:**FIG. 1: EEG ACQUISITION & IMPUTATION SYSTEM ARCHITECTURE**

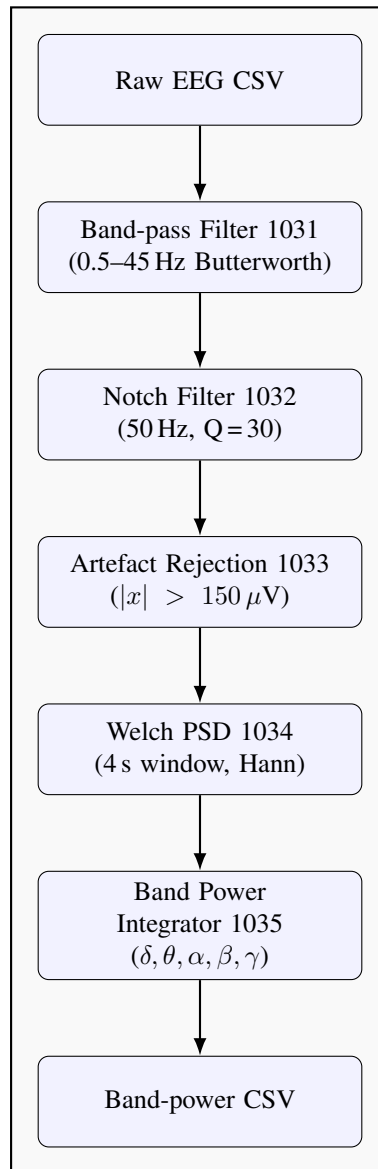
FIG. 2: SIGNAL PROCESSING & FEATURE EXTRACTION

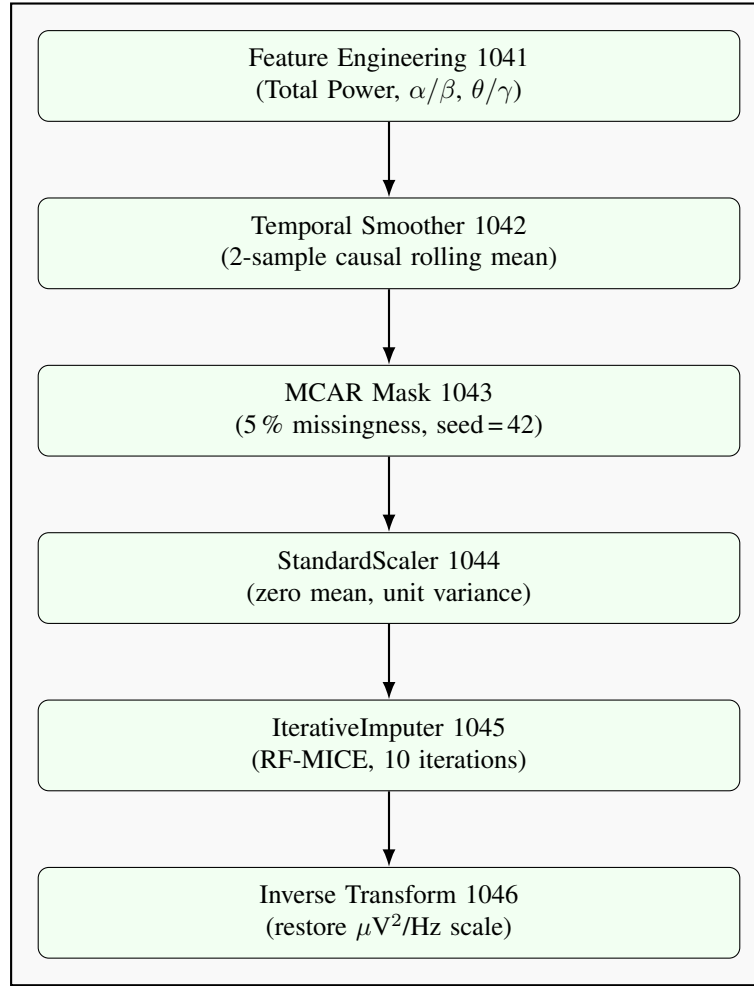
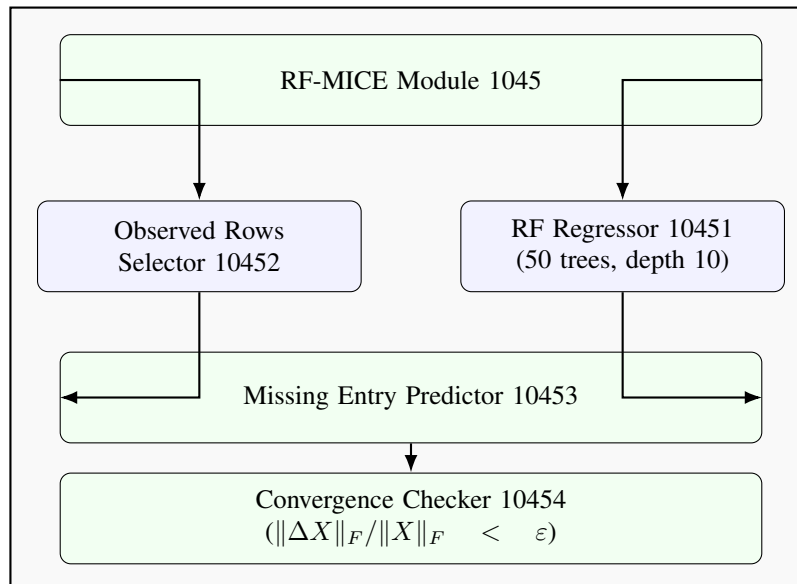
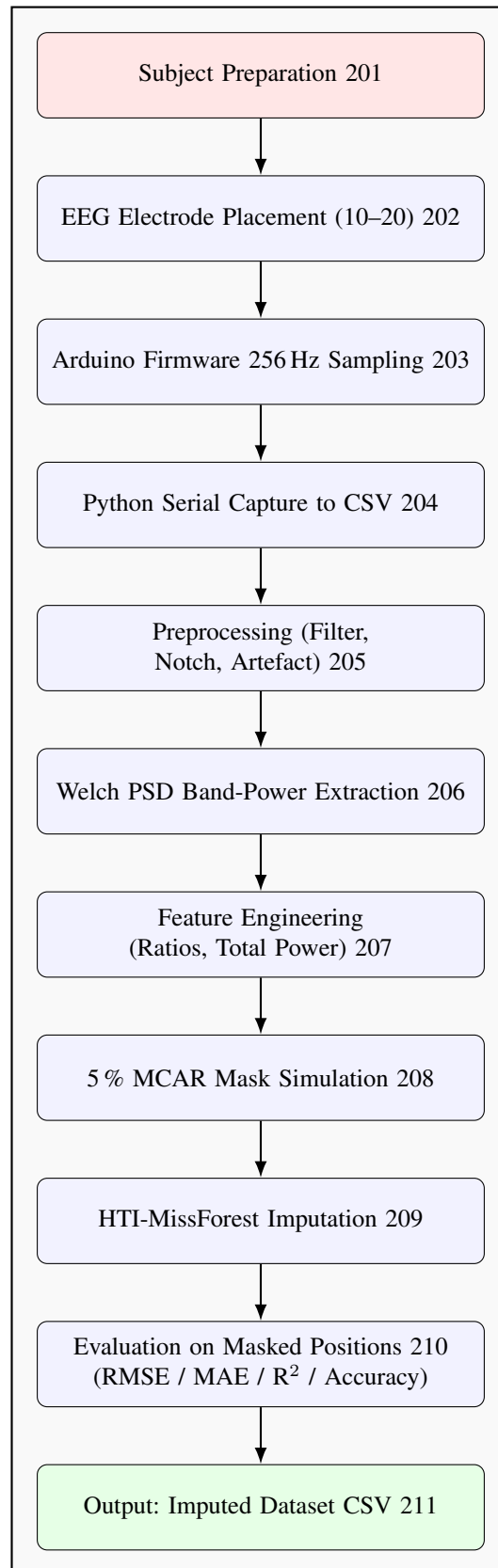
FIG. 3: HTI-MISSFOREST IMPUTATION PIPELINE**FIG. 4: RANDOM FOREST MICE MODULE**

FIG. 5: END-TO-END EEG ACQUISITION, PROCESSING & IMPUTATION

9% Overall Similarity

The combined total of all matches, including overlapping sources, for each database.

Exclusions

▸ 3 Excluded Matches

Match Groups

-  **28 Not Cited or Quoted 9%**
Matches with neither in-text citation nor quotation marks
-  **0 Missing Quotations 0%**
Matches that are still very similar to source material
-  **0 Missing Citation 0%**
Matches that have quotation marks, but no in-text citation
-  **0 Cited and Quoted 0%**
Matches with in-text citation present, but no quotation marks

Top Sources

- 6%  Internet sources
- 1%  Publications
- 7%  Submitted works (Student Papers)

Integrity Flags

0 Integrity Flags for Review

Our system's algorithms look deeply at a document for any inconsistencies that would set it apart from a normal submission. If we notice something strange, we flag it for you to review.

A Flag is not necessarily an indicator of a problem. However, we'd recommend you focus your attention there for further review.

Match Groups

- 28 Not Cited or Quoted 9%**
Matches with neither in-text citation nor quotation marks
- 0 Missing Quotations 0%**
Matches that are still very similar to source material
- 0 Missing Citation 0%**
Matches that have quotation marks, but no in-text citation
- 0 Cited and Quoted 0%**
Matches with in-text citation present, but no quotation marks

Top Sources

- 6% Internet sources
- 1% Publications
- 7% Submitted works (Student Papers)

Top Sources

The sources with the highest number of matches within the submission. Overlapping sources will not be displayed.

1	Student papers	National Institute of Technology, Rourkela on 2025-10-10	3%
2	Internet	www.iiests.ac.in	<1%
3	Student papers	Jawaharlal Nehru University (JNU) on 2018-03-23	<1%
4	Internet	website.nitrkl.ac.in	<1%
5	Internet	allindianpatents.com	<1%
6	Internet	docshare.tips	<1%
7	Publication	Yuji Nishioka, Tatsuya Kokabu, Manami Hada, Rinna Kono, Akihiro Matsuura. "Co...	<1%
8	Internet	patentimages.storage.googleapis.com	<1%
9	Internet	www.nitap.ac.in	<1%
10	Student papers	The University of Texas at Arlington on 2023-08-16	<1%

11	Student papers	Sim University on 2006-10-29	<1%
12	Student papers	University of Exeter on 2024-03-28	<1%
13	Student papers	Indian Institute of Technology Guwahati on 2022-02-10	<1%
14	Publication	Ganesh R. Naik. "Time-Frequency Analysis in Biomedical Engineering - Contempo...	<1%
15	Internet	bitmesra.ac.in	<1%
16	Internet	www.prnewswire.com	<1%
17	Internet	www.reportsanddata.com	<1%
18	Internet	www.sric.iitkgp.ac.in	<1%